Problem Set 15

Products of Powers of Trigonometric Functions

October 15, 2026

Problem 1

(a)

$$\int \sin^3 x \cos^2 x \, dx = \int (\cos^2 x - \cos^4 x) \sin x \, dx$$
$$= \int u^4 - u^2 \, du = \frac{u^5}{5} - \frac{u^3}{3} + C = \boxed{\frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C}$$

(b)

$$u = 3x \implies \frac{1}{3}du = dx$$
$$\int \sin^4 3x \cos^3 3x \, dx = \frac{1}{3}\int \sin^4 u \cos^3 u \, du$$
$$= \frac{1}{3}\int (\sin^4 u - \sin^6 u) \cos u \, du$$
$$= \frac{1}{3}\int v^4 - v^6 \, dv = \frac{1}{3}(\frac{v^5}{5} - \frac{v^7}{7}) + C$$
$$= \boxed{\frac{1}{15}\sin^5 3x - \frac{1}{21}\sin^7 3x + C}$$

(c)

$$\int \sin^2 5x \cos^2 5x \, dx$$
$$= \frac{1}{4}\int (1 - \cos 10x)(1 + \cos 10x) \, dx$$
$$= \frac{1}{4}\int 1 - \cos^2 10x \, dx$$
$$= \frac{1}{4}\int 1 - \frac{1}{2} - \frac{\cos 20x}{2} \, dx$$
$$= \frac{1}{8}(x - \frac{\sin 20x}{20}) + C$$
$$= \boxed{\frac{1}{8}x - \frac{1}{160}\sin 20x + C}$$

(d)

$$\int \cos^5 x\,dx = \int (1 - 2\sin^2 x + \sin^4 x)\cos x\,dx$$

$$= \int u^4 - 2u^2 + 1\,du = \boxed{\frac{1}{5}\sin^5 x - \frac{2}{3}\sin^3 x + \sin x + C}$$

Problem 2

(a)

$$f_{ave} = \frac{\int_{-\pi}^{\pi} \sin^2 x \cos^3 x\,dx}{2\pi}$$

$$\int_{-\pi}^{\pi} \sin^2 x \cos^3 x\,dx = 2\int_0^{\pi} \sin^2 x \cos^3 x\,dx$$

$$= 2\int_0^{\pi} (\sin^2 x - \sin^4 x)\cos x\,dx = 2\int_0^0 u^2 - u^4\,du = 0$$

$$f_{ave} = \boxed{0}$$

$$u = \pi - x \implies \int_0^{\pi} \sin^2 u \cos^3 u\,du = -\int_{-\pi}^0 \sin^2 u \cos^3 u\,du$$

(b)

$$\int_{-\pi}^{\pi} \sin^2 x \cos^2 x\,dx = 2\int_0^{\pi} \sin^2 x \cos^2 x\,dx$$

$$= \frac{1}{2}\int_0^{\pi} (1 - \cos 2x)(1 + \cos 2x)\,dx$$

$$= \frac{1}{2}\int_0^{\pi} 1 - \cos^2 2x\,dx$$

$$= \frac{1}{2}\int_0^{\pi} 1 - \frac{1}{2} - \frac{1}{2}\cos 4x\,dx$$

$$= \frac{1}{2}(\frac{1}{2}x - \frac{1}{8}\sin 4x)\Big|_0^{\pi} = \frac{\pi}{4}$$

$$g_{ave} = \frac{\pi}{4} \cdot \frac{1}{2\pi} = \boxed{\frac{1}{8}}$$

Problem 3

(a)

$$s(t) = \int \sin(\omega t)\cos^2(\omega t)\,dt$$

$$= \begin{bmatrix} u = \cos \omega t \\ du = -\omega \sin \omega t\,dt \end{bmatrix} - \frac{1}{\omega}\int u^2\,du$$

$$= \boxed{-\frac{\cos^3(\omega t)}{3\omega} + \frac{1}{3\omega}}$$

(b)

$$-\frac{\cos^3(\omega t)}{3\omega} \in [-\frac{1}{3\omega}, \frac{1}{3\omega}]$$
$$\iff -\frac{\cos^3(\omega t)}{3\omega} + \frac{1}{3\omega} \in [0, \frac{2}{3\omega}]$$
$$\cos^3(\omega t) = -1 \implies \boxed{t = \frac{\pi}{\omega}}$$

(c) Yes, it is periodic.

$$t = \frac{2\pi}{\omega}$$
$$s(\frac{\pi}{\omega}) - s(0) = \frac{2}{3\omega}$$
$$2 \cdot \frac{2}{3\omega} = \boxed{\frac{4}{3\omega}}$$

Problem 4

(a)

$$\int \tan^3 x \sec^3 x \, dx = \int \sec^2 x(\sec^2 x - 1) \sec x \tan x \, dx$$
$$= \int u^4 - u^2 \, du = \frac{u^5}{5} - \frac{u^3}{3} + C$$
$$= \boxed{\frac{1}{5}\sec^5 x - \frac{1}{3}\sec^3 x + C}$$

(b)

$$\int \tan^4 x \sec^4 x \, dx = \int \tan^4 x(1 + \tan^2 x) \sec^2 x \, dx$$
$$= \int u^4 + u^6 \, du = \boxed{\frac{1}{7}\tan^7 x + \frac{1}{5}\tan^5 x + C}$$

(c)

$$\int \tan^2 x \sec x \, dx = \int \sec^3 x - \sec x \, dx$$
$$= \boxed{\frac{1}{2}\sec x \tan x - \frac{1}{2}\ln|\sec x + \tan x| + C}$$

Problem 5

(a)

$$\int \sec^n x \, dx = \begin{bmatrix} u = \sec^{n-2} x & dv = \sec^2 x \\ du = (n-2)\sec^{n-2} x \tan x \, dx & v = \tan x \end{bmatrix}$$

$$\begin{aligned}
&\sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \tan^2 x \, dx \\
&= \sec^{n-2} x \tan x - (n-2) \int \sec^n x \, dx + (n-2) \int \sec^{n-2} x \, dx \\
&= \boxed{\frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx}
\end{aligned}$$

(b)

$$\int \sec^3 x \, dx = \boxed{\frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C}$$

(c)

$$\begin{aligned}
\int \sec^5 x \, dx &= \frac{\sec^3 x \tan x}{4} + \frac{3}{4}(\frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x|) + C \\
&= \boxed{\frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + C}
\end{aligned}$$

(d)

$$\begin{aligned}
\int \tan^2 x \sec^3 x \, dx &= \int \sec^5 x - \sec^3 x \, dx \\
&= \boxed{\frac{1}{4} \sec^3 x \tan x - \frac{1}{8} \sec x \tan x - \frac{1}{8} \ln |\sec x + \tan x| + C}
\end{aligned}$$

(e)

$$\int \sec^4 x \, dx = \boxed{\frac{1}{3} \sec^2 x \tan x + \frac{2}{3} \tan x + C}$$

Problem 6

(a)

$$\begin{aligned}
\int \frac{\cos^3 x}{\sqrt{\sin x}} \, dx &= \int \frac{1 - \sin^2 x}{\sqrt{\sin x}} \cos x \, dx \\
&= \int \frac{1}{\sqrt{u}} - u^{\frac{3}{2}} \, du = 2\sqrt{u} - \frac{2}{5} u^{\frac{5}{2}} + C \\
&= \boxed{2\sqrt{\sin x} - \frac{2}{5} \sin^{\frac{5}{2}} x + C}
\end{aligned}$$

(b)

$$\begin{aligned}
\int \frac{\sin^2 x}{\cos^4 x} \, dx &= \int \sec^2 x \tan^2 x \, dx \\
&= \boxed{\frac{1}{3} \tan^3 x + C}
\end{aligned}$$

(c)

$$\int \cot^3 x \csc^3 x \, dx = \int (\csc^4 x - \csc^2 x) \cot x \csc x \, dx$$
$$= \boxed{\frac{1}{3}\csc^3 x - \frac{1}{5}\csc^5 x + C}$$

(d)

$$\int \tan^3 x \, dx = \int \frac{\sec^2 x - 1}{\sec x} \sec x \tan x \, dx$$
$$= \boxed{\frac{1}{2}\sec^2 x - \ln|\sec x| + C}$$

Problem 7

(a)

$$\int_{-\pi}^{\pi} \cos^2(nx) \, dx = 2\int_0^{\pi} \cos^2 nx \, dx$$
$$= \int_0^{\pi} 1 + \cos 2nx \, dx = x + \frac{1}{2n}\sin 2nx\Big|_0^{\pi} = \boxed{\pi}$$

(b)

$$E(a) = \int_{-\pi}^{\pi} (f(x) - a\cos(nx))^2 \, dx$$
$$= \int_{-\pi}^{\pi} f(x)^2 - 2a\cos(nx)f(x) + a^2\cos(nx)^2 \, dx$$
$$= a^2\int_{-\pi}^{\pi} \cos(nx)^2 \, dx - 2a\int_{-\pi}^{\pi} \cos(nx)f(x) \, dx + \int_{-\pi}^{\pi} f(x)^2 \, dx$$
$$= \pi a^2 - 2a\int_{-\pi}^{\pi} \cos(nx)f(x) \, dx + \int_{-\pi}^{\pi} f(x)^2 \, dx$$
$$a = \boxed{\frac{1}{\pi}\int_{-\pi}^{\pi} \cos(nx)f(x) \, dx}$$

(c)

$$a_1 = \boxed{-4}$$
$$a_2 = \boxed{1}$$

(d)

$$a = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos(nx) \, dx = \frac{2}{\pi}\int_0^{\pi} x^2\cos(nx) \, dx$$

$$= \begin{bmatrix} u = x^2 & dv = \cos nx\,dx \\ du = 2x\,dx & v = \frac{1}{n}\sin nx\,dx \end{bmatrix} \frac{2}{n\pi}x^2 \sin nx\Big|_0^\pi - \frac{4}{n\pi}\int_0^\pi x \sin nx\,dx$$
$$= \begin{bmatrix} u = x & dv = \sin nx\,dx \\ du = dx & v = -\frac{1}{n}\cos nx \end{bmatrix} + \frac{4}{n^2\pi}x\cos nx\Big|_0^\pi - \frac{4}{n^2\pi}\int_0^\pi \cos nx\,dx$$
$$= \boxed{\frac{4(-1)^n}{n^2}}$$

The equation is a great approximation of the function $y = x^2$ with cosine.

Problem 8

(a)

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x \cos(nx)\,dx$$
$$= \begin{bmatrix} u = \cos^n x & dv = \cos(nx)\,dx \\ du = -n\cos^{n-1}x \sin x & u = \frac{1}{n}\sin(nx) \end{bmatrix} \frac{1}{n}\cos^n x \sin nx\Big|_0^{\frac{\pi}{2}}$$
$$+ \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \sin(nx)\,dx = \boxed{\int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \sin(nx)\,dx}$$

(b)

$$\cos((n-1)x) = \cos nx \cos x + \sin nx \sin x$$
$$2I_n = \int_0^{\frac{\pi}{2}} \cos^n x \cos(nx)\,dx + \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \sin(nx)\,dx$$
$$= \int_0^{\frac{\pi}{2}} \cos^{n-1} x(\cos nx \cos x + \sin nx \sin x)\,dx$$
$$= \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cos((n-1)x)\,dx = I_{n-1}$$
$$I_n = \frac{1}{2}I_{n-1}, I_0 = \frac{\pi}{2} \implies \boxed{I_n = \frac{\pi}{2^{n+1}}}$$

(c)

$$K_n = \int_0^{\frac{\pi}{2}} \cos^n x \sin(nx)\,dx$$
$$= \begin{bmatrix} u = \cos^n x & dv = \sin(nx)\,dx \\ du = -n\cos^{n-1}x \sin x & v = -\frac{1}{n}\cos(nx) \end{bmatrix} - \frac{1}{n}\cos^n x \cos nx\Big|_0^{\frac{\pi}{2}}$$
$$+ \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \cos(nx)\,dx = \frac{1}{n} - \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \cos(nx)\,dx$$
$$\sin((n-1)x) = \sin nx \cos x - \cos nx \sin x$$
$$K_{n-1} = \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin((n-1)x)\,dx$$

$$= \int_0^{\frac{\pi}{2}} \cos^{n-1} x(\sin nx \cos x - \cos nx \sin x)\, dx$$

$$= \int_0^{\frac{\pi}{2}} \cos^n x \sin nx\, dx - \int_0^{\frac{\pi}{2}} \cos^{n-1} x \sin x \cos nx\, dx = 2K_n - \frac{1}{n}$$

$$\boxed{K_n = \frac{1}{2}K_{n-1} + \frac{1}{2n}}$$

$$K_n = \frac{1}{2}K_{n-1} + \frac{1}{2n} = \frac{1}{4}K_{n-2} + \frac{1}{2n} + \frac{1}{4(n-1)}$$

$$= \cdots = \left(\frac{1}{2}\right)^n K_0 + \sum_{k=1}^{n} \left(\frac{1}{2}\right)^{n-k+1} \frac{1}{k}$$

$$\boxed{K_n = \frac{1}{2^{n+1}} \sum_{k=1}^{n} \frac{2^k}{k}}$$

(d)

$$I_1 = \frac{\pi}{4}, I_2 = \frac{\pi}{8}$$
$$K_1 = \frac{1}{2}, K_2 = \frac{1}{2}$$